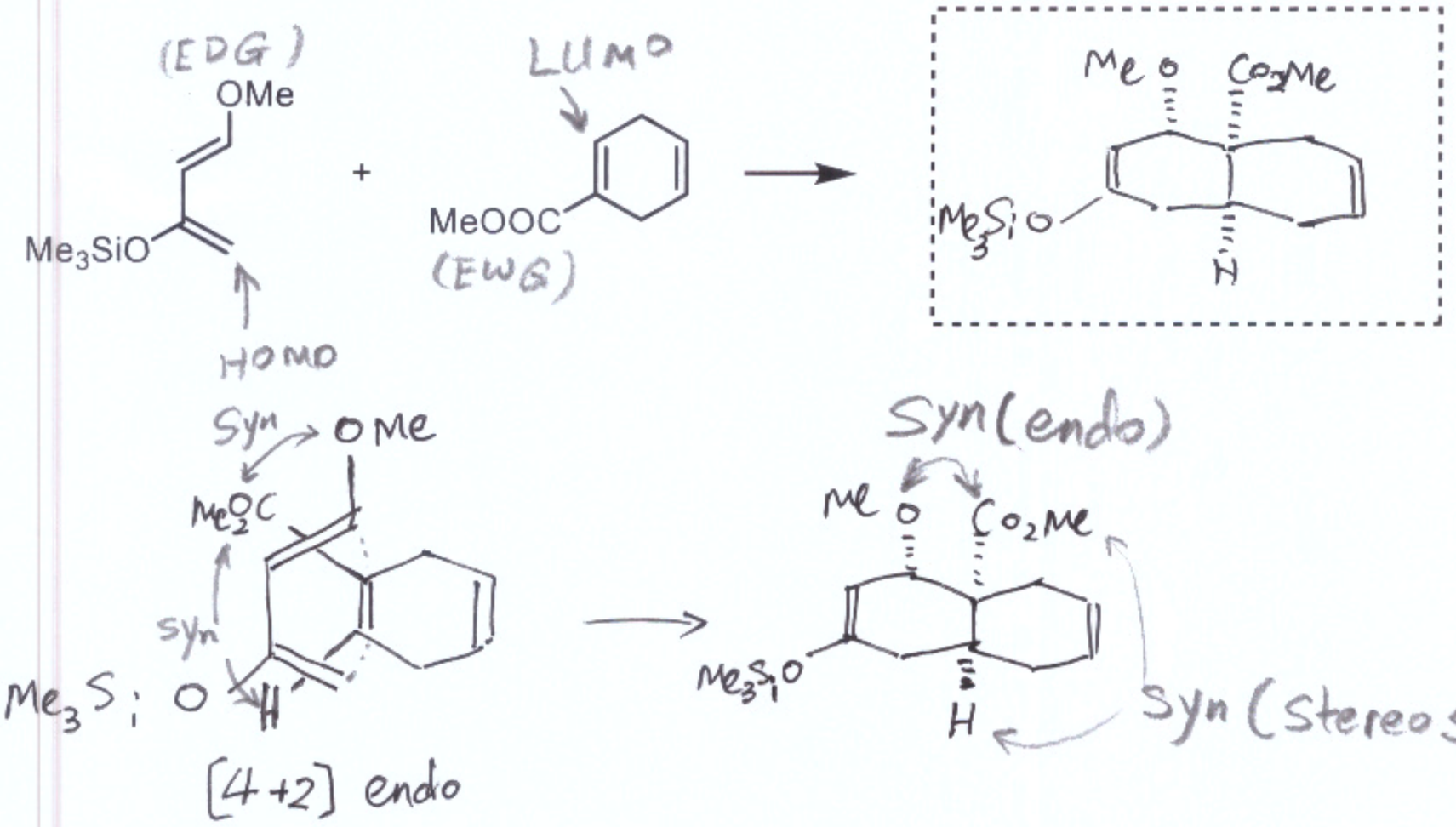
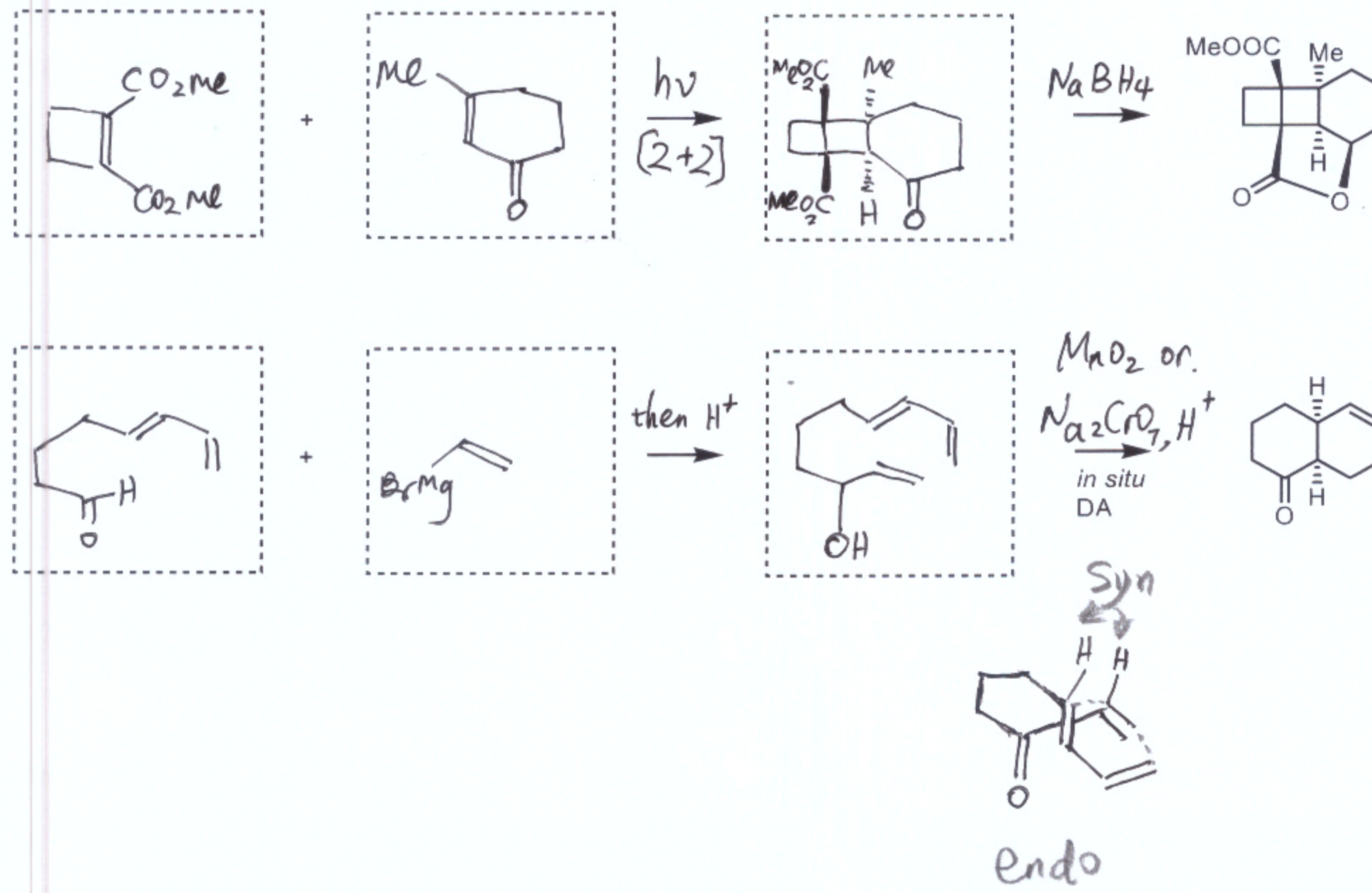
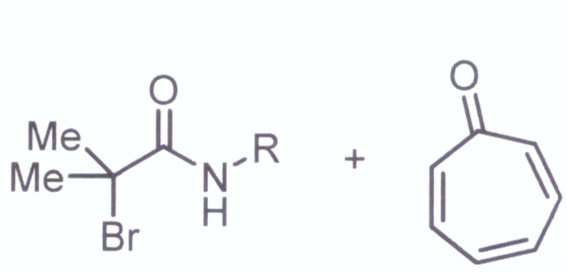
Question 1. Proposed the structures of pericyclic reaction products. Explain the regio- and stereo-selectivity of each reaction if applicable.



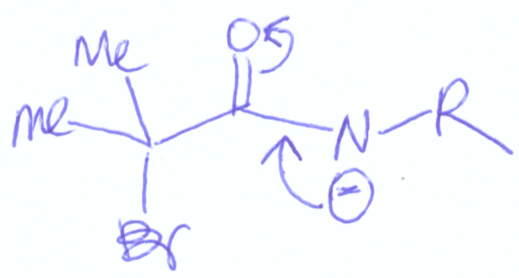
Question 2. Propose the reactants and reagents for the synthesis of the target molecules. Explain the regio- and stereo- selectivity of each reaction if applicable.

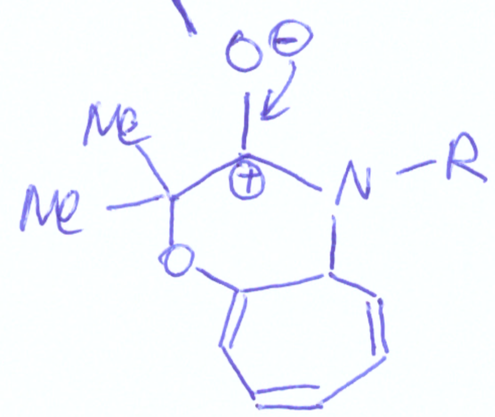


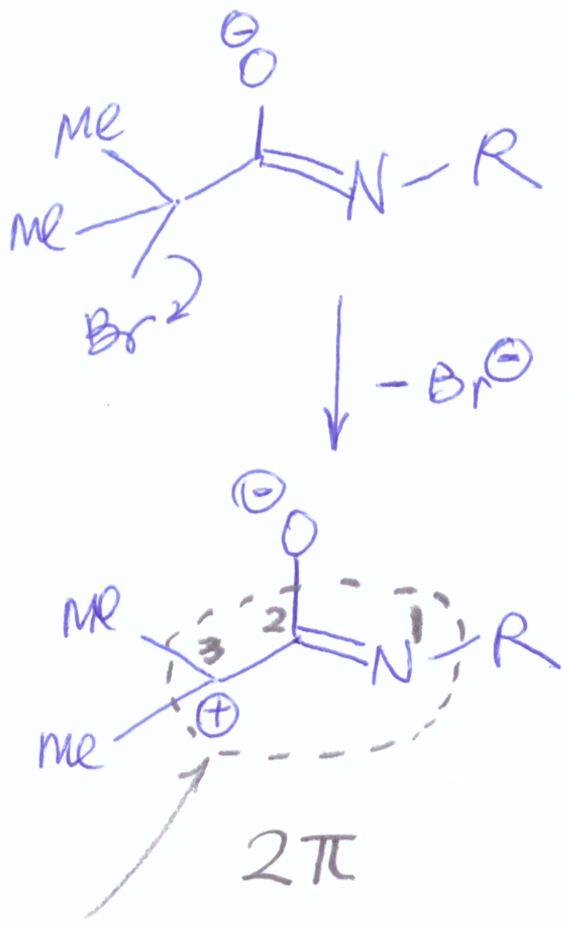
Question 3. Propose a reaction mechanism for the following reaction (Hint: the key step is a cycloaddition reaction). Explain the regioselectivity. Is this reaction allowed according to Woodward-Hoffmann rules?

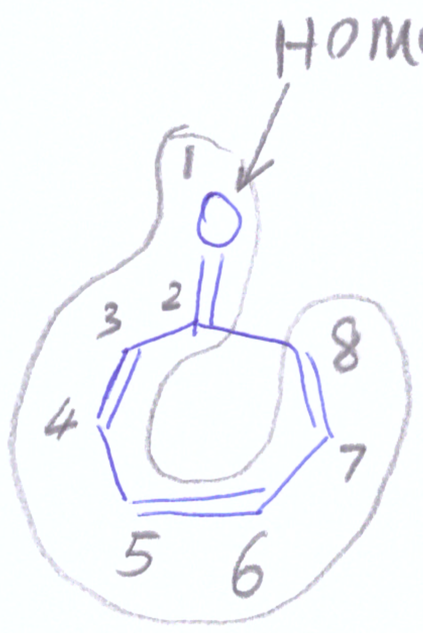


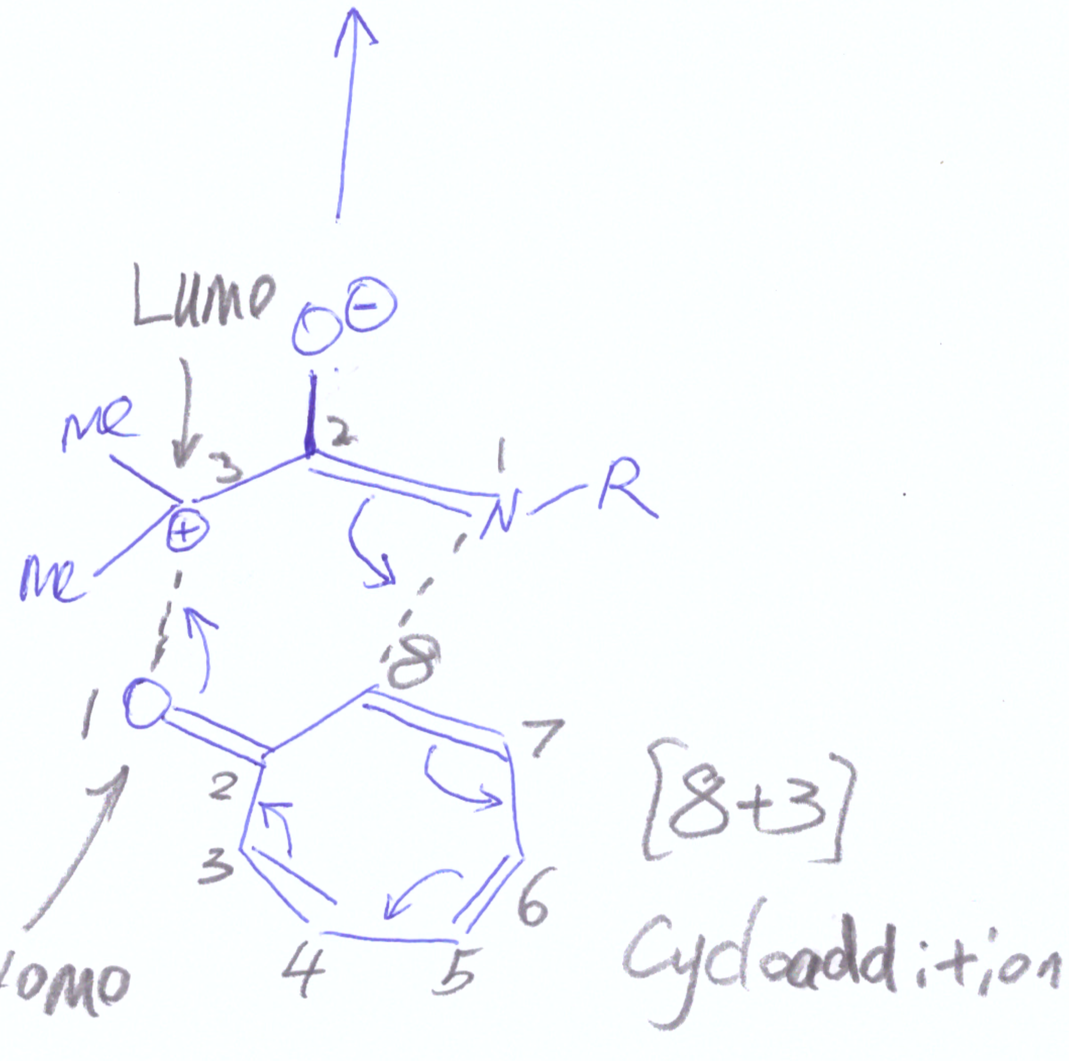
$-H^+$ ↓ Na_2CO_3











$2\pi_{(s)} + 8\pi_{(s)}$

↑ Counted ↑ Not Counted

In total: 1 component (Allowed)